When Do Benchmarks Stop Ranking Models? Item Count and Cohort Spread, Not Difficulty Shape, Set the Saturation Point

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Abstract

Leaderboards rank models long after their benchmarks have stopped measuring real differences. A common belief is that the shape of a benchmark's item-difficulty distribution decides when this happens: uniform benchmarks should saturate near 85% accuracy, heavy-tailed ones should last until 95%, a benchmark's useful lifetime should track the variance of its item difficulties, and at saturation evaluation-seed noise should outweigh capability in rankings. We test these claims with a matched-variance item response theory (IRT) analysis, per-item responses of 220 Open LLM Leaderboard v1 models on 62 tasks, real items resampled into prescribed difficulty shapes, and 225,000 option-order-permuted evaluations of 15 open LLMs. Shape barely matters. With variance held fixed, all seven shapes lose discrimination within about 1.5 points of each other (95.2–96.6% accuracy for 500 items), and the hard-tailed shape fails first, not last. Real items reproduce this null. The saturation accuracy is instead set by the number of items (80–84% at 100 items, 96% at 500, 99% at 2000) and by how tightly the competing models cluster. Difficulty variance predicts lifetime only among equal-size benchmarks (r=0.92 in simulation at $N \geq 500$), and not across real tasks (r=−0.06, 95% CI $[-0.38, 0.33]$). Seed noise is about 0.40× the binomial item-sampling noise. It explains at most 18% of score variance on 500-item benchmarks and dominates only for near-tied models on about 150 items at ≥99% accuracy. Benchmark builders should spend items near the frontier and keep test sets large; difficulty-histogram shape is a second-order concern.

1 Introduction

Benchmarks saturate quickly. A new test set is released, frontier models climb toward its ceiling within a few years, and soon the top of the leaderboard is a cluster of models separated by fractions of a point [Kiela et al., 2021, Ott et al., 2022]. Those small gaps still drive model selection, paper acceptances, and research attention. **When do they stop meaning anything?** If we could predict this point from the design of a benchmark, builders could make benchmarks that last longer, and readers would know when to stop trusting a rank difference.

A natural candidate is the *item-difficulty distribution*. Item response theory (IRT) tells us that an item discriminates best between models whose ability is close to its difficulty [Lord, 1980, Maia Polo et al., 2024]. A benchmark whose items all sit at one difficulty level should therefore stop separating models once they pass that level, while a benchmark with a long tail of hard items should keep a reserve of informative items for stronger models. This intuition leads to a concrete hypothesis that we test in this paper:

- **H1 (shape).** Benchmarks with uniform difficulty distributions lose discriminative power once cohort mean accuracy exceeds about 85%; heavy-tailed distributions keep discriminating up to about 95%.

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- **H2 (noise).** At saturation, evaluation-seed variation accounts for more ranking variance than true capability differences.
- **H3 (lifetime).** A benchmark's effective lifetime correlates strongly ($r > 0.8$) with the variance of its item difficulties.

Prior work has established the pieces but not the link between them. IRT analyses show that LLM benchmarks have strong latent structure and that few items carry most of the information [Maia Polo et al., 2024, Kipnis et al., 2024, Rodriguez et al., 2021]. Separate work shows that evaluation scores carry non-trivial seed and prompt variance [Madaan et al., 2024, Miller, 2024, Zheng et al., 2024]. What is missing is a quantitative account of *which* benchmark property decides the accuracy at which discrimination collapses, one that separates the shape of the difficulty distribution from its variance and from the number of items, together with a measurement of how seed noise compares with capability differences as a benchmark saturates.

We answer this with four complementary experiments. First, we build a controlled IRT model in which we vary the shape of the difficulty distribution (seven shapes) while holding its variance fixed, and we compute exactly when the ranking of a cohort of similar models becomes dominated by noise. Second, we test the same predictions on real per-item responses of 220 Open LLM Leaderboard v1 models on 62 tasks, including benchmarks we build by resampling real MMLU and HELLASWAG items into prescribed difficulty shapes. Third, we run 15 open LLMs with 10 option-order seeds on 1,500 items to measure seed noise directly. Fourth, we track the real frontier cohort of the leaderboard month by month.

The evidence goes against two parts of the hypothesis and gives only conditional support to the third. With variance matched, all seven difficulty shapes lose discrimination within about 1.5 points of each other (at 95.2–96.6% accuracy for a 500-item benchmark), and the hard-tailed shape fails slightly *earlier*. Real items resampled into the same shapes reproduce this null. What moves the saturation point is the *number of items* (80–84% at 100 items, 96% at 500, 99% at 2000) and the *spread of the competing models* (83–88% for a tight cohort, 99% for a wide one). Difficulty variance predicts lifetime only at fixed size ($r = 0.92$ at $N \geq 500$); the correlation drops to $r = 0.20$ when sizes vary, turns negative at $N = 100$, and is absent across real tasks ($r = -0.06$). Seed noise is about $0.40\times$ the binomial item-sampling noise, explains 1–18% of score variance on 500-item benchmarks, and dominates only for near-tied models on small subsets at $\geq 99\%$ accuracy. A short calculation (section 5.1) explains all three findings: at fixed accuracy, the error mass $1 - A$ caps the number of informative items, and shape only decides how much of that capped budget is wasted.

Our contributions are:

- We separate the effects of difficulty shape, difficulty variance, benchmark size, and cohort spread on the accuracy at which a benchmark stops discriminating, and show that size and cohort spread dominate while shape changes the saturation point by about 1.5 points.
- We test the hypotheses on real data from 220 leaderboard models and 62 tasks, including a direct test with real items resampled into prescribed difficulty shapes, and find no relation between difficulty variance and lifetime across real tasks.
- We measure evaluation-seed noise for 15 open LLMs with 10 option-order seeds each, and map when it overtakes capability differences as a function of accuracy and item count.
- We give a simple approximation that explains these results and translate it into concrete guidance: budget items near the frontier, keep N large, and report seed-level error bars for near-tied models on small, near-ceiling test sets.

Section 2 reviews related work. Section 3 defines our measures and experiments. Section 4 reports results for each hypothesis, and section 5 discusses the mechanism, implications, and limitations.

2 Related Work

Benchmark saturation and dynamics. Benchmarks in NLP and machine learning are saturated faster than they are created. Kiela et al. [2021] document how quickly models reach human-level scores on static benchmarks and propose dynamic, adversarial data collection as a remedy. Ott et al. [2022] map the life cycle of thousands of benchmarks and show that many saturate within a few years or stop being used. Bowman and Dahl [2021] argue that benchmarks must keep being valid and discriminative near the top of the range to remain useful. These works describe *that* benchmarks

saturate. We ask *which design properties* decide the accuracy at which a benchmark stops separating contemporaneous models.

Item response theory for model evaluation. IRT models the probability that a respondent solves an item as a function of ability and item difficulty [Rasch, 1960, Lord, 1980]. Lalor et al. [2016] and Vania et al. [2021] brought IRT to NLP test sets, and Rodriguez et al. [2021] showed that items are not equally informative for leaderboard ranking. More recently, tinyBenchmarks [Maia Polo et al., 2024] and metabench [Kipnis et al., 2024] use IRT to compress LLM benchmarks to a small, informative core, and adaptive testing has been proposed for LLMs [Zhuang et al., 2023]. Building on this line, we use IRT not to compress benchmarks but to ask how the difficulty distribution controls the saturation point, and we check every IRT-based claim on real LLM responses.

Variance and noise in LLM evaluation. Madaan et al. [2024] quantify seed variance across training runs and show that it can rival the gaps between models on several benchmarks. Miller [2024] argues for reporting standard errors based on item sampling. Multiple-choice evaluations are also sensitive to option order [Zheng et al., 2024, Pezeshkpour and Hruschka, 2024] and to prompt formatting [Sclar et al., 2024]. Holistic evaluation suites report many such factors side by side [Liang et al., 2023]. Unlike these works, we measure evaluation-seed noise as a function of how saturated the benchmark is, and we compare it directly with the capability spread of the models being ranked, so that we can say when seed noise dominates and when it does not.

3 Methods

Our main question is: **does the shape or variance of the item-difficulty distribution decide the mean accuracy at which a benchmark stops discriminating between models, and does evaluation-seed noise dominate rankings at that point?** We first define the quantities we measure (section 3.1), then describe four experiments (section 3.2–section 3.5). Exp 1 is a controlled IRT model; Exp 2–4 use real LLM responses and re-test every claim from Exp 1.

3.1 Definitions

Cohort. A cohort is a set of K contemporaneous models of similar ability, standing for the current frontier. In the IRT model, cohort abilities are $\theta_k \sim \mathrm{Normal}(\mu, \sigma_c^2)$; moving the cohort centre μ upward stands in for progress over time.

Reliability. We measure discrimination by the reliability ρ, the share of between-model score variance that is *not* noise. Let $\bar{s}_k$ be the expected accuracy of model k on an N-item benchmark and σ_k^2 the variance of its observed accuracy. Then

$$\rho = \frac{\mathrm{Var}_k(\bar{s}_k)}{\mathrm{Var}_k(\bar{s}_k) + \frac{1}{K}\sum_k \sigma_k^2}. \tag{1}$$

Thus $1 - \rho$ is the noise share of ranking variance, and $\rho < 0.5$ means noise outweighs capability. We use $\rho < 0.5$ as the threshold for "lost discriminative power". On real data we estimate ρ either by split-half reliability across the cohort (Spearman–Brown corrected) or as $1 -$ binomial noise/observed variance.

Saturation accuracy and lifetime. The saturation accuracy $A^\star$ is the cohort mean accuracy at which ρ first falls below 0.5 as the cohort's ability rises. We also report thresholds of $\rho = 0.7$ and Kendall $\tau = 0.5$ between observed and true rankings. The effective lifetime is the length of the ability axis over which $\rho \geq 0.5$; under steady progress this is proportional to time.

3.2 Exp 1: Controlled IRT model

Response model. Model k answers item i correctly with probability

$$P_{ik} = c + (1 - c)\,\sigma\big(a_i(\theta_k - b_i)\big), \qquad a_i \sim \mathrm{LogNormal}(0, 0.3^2), \tag{2}$$

with guessing floor $c = 0$ unless stated otherwise. We stress that this simulates item *responses*; it is not a simulation of LLMs.

Difficulty shapes. We draw difficulties b_i from seven shapes with standard deviation matched to 1.5 logits: point mass (all items equally hard), uniform, normal, Laplace, Student-t_3, a log-normal with a long *hard* tail (lognormal-hard), and its mirror image with a long *easy* tail (lognormal-easy).

Protocol. The base setting uses $N = 500$ items, $K = 20$ models and $\sigma_c = 0.3$ logits. We sweep the cohort centre over 221 ability levels. Because the score of model k is a sum of independent Bernoulli variables, we compute its mean and variance exactly (Poisson-binomial), which gives ρ in equation 1 without sampling noise; Kendall τ is estimated with 200 Monte Carlo replicates. Each shape is replicated with 20 independent item draws. We repeat the analysis under ten sensitivity settings: $N \in \{100, 2000\}$, $\sigma_c \in \{0.15, 0.6\}$, difficulty SD $\in \{0.75, 3\}$, guessing $c = 0.25$, $K = 10$, and fixed $a_i = 1$. To test H3, we draw 600 random benchmark configurations (random shape, SD in $[0.25, 4]$, $N \in \{100, 250, 500, 1000\}$) and correlate their lifetime with $\mathrm{Var}(b)$.

3.3 Exp 2: Real leaderboard responses

Data. We download per-item correctness from the public Open LLM Leaderboard v1 "details" datasets [Beeching et al., 2023]. We sample 245 models stratified across 20 quantile bins of the leaderboard average plus the top 25; the 220 models with results on all 62 tasks are used. The tasks are ARC-CHALLENGE [Clark et al., 2018] (25-shot), HELLASWAG [Zellers et al., 2019] (10-shot), TRUTHFULQA-mc1 [Lin et al., 2022], WINOGRANDE [Sakaguchi et al., 2020] (5-shot), GSM8K [Cobbe et al., 2021] (5-shot), and the 57 MMLU subjects [Hendrycks et al., 2021] (5-shot), with 91–10,019 items per task. We key items by a hash of their normalised text and keep items answered by at least 95% of models, which leaves no missing cells.

Calibration/evaluation split. To avoid selection leakage, we order models by an *external* ability score (the mean z-score on all benchmarks except the task under study) and assign alternate models to a calibration half and an evaluation half (about 110 each). We estimate item difficulties only on the calibration half and measure discrimination only on the evaluation half.

Analyses. (a) We fit a Rasch model per task (joint MAP with an $\mathrm{Normal}(0, 3^2)$ prior, L-BFGS) and record $\mathrm{Var}(b)$, SD, IQR, 90% range, skew, and kurtosis. We first tried a fixed 25% guessing floor and rejected it: 26% of ARC-CHALLENGE items are answered *below* chance by the model population, which a guessing floor cannot fit. (b) We slide cohorts of $K = 20$ evaluation models along the external ability axis (step 2) and compute split-half reliability on random 90-item subsamples, so all tasks have equal N. The *empirical lifetime* is the fraction of cohorts with $\rho \geq 0.5$. (c) A *model-based lifetime* evaluates the Exp-1 ρ-curve with each task's real fitted difficulties ($N = 90$, cohort spread estimated from data), in units of the population θ-SD. (d) We correlate both lifetimes with $\mathrm{Var}(b)$ across the 62 tasks (Pearson and Spearman, 2,000-sample bootstrap CIs, partial correlation given mean accuracy). (e) As a direct test of H1, we build 200-item benchmarks from the full MMLU (14k items) and HELLASWAG (10k items) pools by importance resampling to prescribed shapes (uniform, normal, Laplace, t_3, lognormal-hard; SD 1 logit), at 12 difficulty locations with 6 replicates per shape, and score them with the evaluation half's real responses.

3.4 Exp 3: Seed variance of real LLMs

Models and items. We run 15 open instruction-tuned LLMs locally in bf16: Qwen2.5-{0.5, 1.5, 3, 7, 14}B-Instruct [Qwen Team, 2024], Qwen3-{1.7, 4, 8}B with thinking disabled [Yang et al., 2025], SmolLM2-1.7B-Instruct [Ben Allal et al., 2025], Phi-3.5-mini-instruct [Abdin et al., 2024], Phi-4-mini-instruct [Microsoft, 2025], OLMo-2-1124-7B-Instruct [Team OLMo, 2024], granite-3.1-8b-instruct, Yi-1.5-9B-Chat [Young et al., 2024], and Mistral-7B-Instruct-v0.3 [Jiang et al., 2023]. We use 1,500 items: 500 each from the test splits of ARC-EASY, ARC-CHALLENGE, and MMLU.

Prompting and seeds. We use a 0-shot chat-template multiple-choice prompt followed by the assistant prefix "The answer is", and read the answer from the next-token probabilities of the letters A–D, renormalised over these four letters. Each item is evaluated under $S = 10$ seeds: seed 0 keeps the original option order and seeds 1–9 apply random permutations (15,000 forward passes per model). The *greedy* score takes the argmax letter, so only option order varies; the *sampled* score draws the answer from the letter distribution at temperature 1, adding sampling noise.

Variance decomposition. Let σ^2_{seed} be the mean over models of the across-seed variance of accuracy. The capability variance is $\sigma^2_{\text{cap}} = \mathrm{Var}_m(\bar{s}_m) - \sigma^2_{\text{seed}}/S$, where $\bar{s}_m$ is the seed-averaged accuracy of model m. The seed share is $\sigma^2_{\text{seed}}/(\sigma^2_{\text{seed}} + \sigma^2_{\text{cap}})$. We also report Kendall τ between a single-seed ranking and the seed-averaged ranking, the probability that two single-seed runs flip the order of a model pair, and agreement on the top-1 model. We study (i) the natural 500-item benchmarks with cohorts of all 15 models, the top 10, and the top 6; and (ii) difficulty-stratified 150-item subsets (15

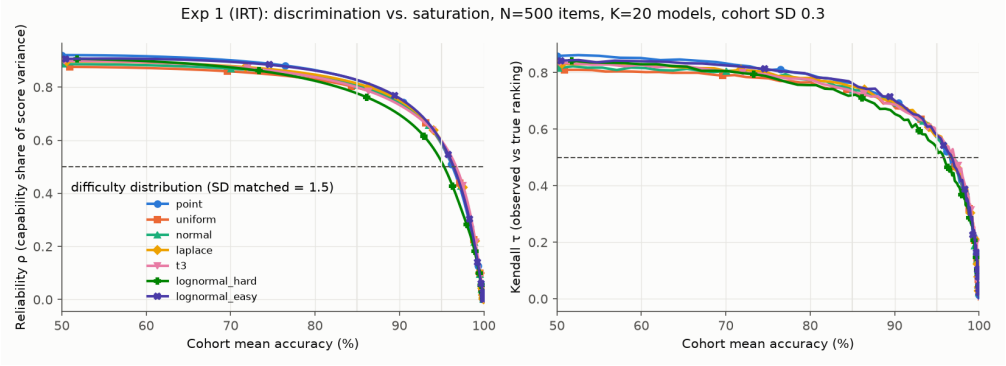


Figure 1: **Discrimination vs. saturation in the IRT model** ($N = 500$ items, $K = 20$ models, cohort SD $\sigma_c = 0.3$, difficulty SD matched at 1.5 logits). *(Left)* Reliability ρ (capability share of score variance). *(Right)* Kendall τ between observed and true rankings. The dashed line marks 0.5. All seven difficulty shapes follow nearly the same curve and cross 0.5 between 95% and 97% accuracy; the hard-tailed log-normal (dark green) crosses first.

Setting	**point**	**uniform**	**normal**	**Laplace**	t_3	**lognormal-hard**	**lognormal-easy**
Base (N=500, σ_c=0.3, SD=1.5)	96.4	**96.6**	96.5	96.2	96.1	95.2	96.5
$N = 100$ items	**84.5**	82.1	80.1	80.6	80.8	79.1	84.4
$N = 2000$ items	**99.0**	**99.0**	**99.0**	**99.0**	98.8	98.5	**99.0**
Tight cohort $\sigma_c = 0.15$	**87.5**	86.4	85.0	84.4	85.0	83.1	**87.5**
Wide cohort $\sigma_c = 0.6$	**99.0**	**99.0**	**99.0**	**99.0**	98.9	98.5	**99.0**
Wide difficulties SD $= 3$	96.4	**96.7**	95.7	94.5	94.8	92.4	96.5
Guessing $c = 0.25$	96.5	**96.6**	96.4	96.0	96.0	94.8	96.5

Table 1: **Saturation accuracy $A^\star$ (cohort mean accuracy, %, at which $\rho < 0.5$) in the IRT model.** Values are means over 20 item draws; difficulty SD is matched within each row. Higher means the benchmark discriminates longer; the latest-saturating shape in each row is in **bold**. Shape moves $A^\star$ by about 1.5 points in the base setting, and the heavy hard tail (lognormal-hard) saturates *earliest* in every row. In contrast, item count (N) and cohort spread (σ_c) move $A^\star$ by 15–20 points.

windows from hardest to easiest), where item difficulty is estimated on the *weaker* 7 models and seed share is measured on the *stronger* 8, so that selection does not leak into the measurement.

Cross-check. We fit the per-item seed-noise constant $\kappa = N \cdot \mathrm{Var}_{\text{seed}}(\text{acc}) \approx r \cdot \text{acc}(1 - \text{acc})$ and apply it to the score spreads of the real leaderboard top-20.

3.5 Exp 4: Leaderboard history

For each month from September 2023 to June 2024, we take the top 20 unique models by leaderboard average among all submissions up to that month (over 7,000 entries) and compute, for each of the six benchmarks, the cohort mean, SD, binomial noise, and ρ.

Implementation. All analyses use Python with NumPy, SciPy, statsmodels and PyTorch; random seed 42 throughout. Exp 1, 2 and 4 run on a 32-core CPU; Exp 3 runs on two RTX A6000 GPUs (1–6 minutes per model). No paid API is used.

4 Results

We report results by hypothesis: difficulty shape (H1, section 4.1), difficulty variance and lifetime (H3, section 4.2), and seed noise (H2, section 4.3). We close with the real leaderboard trajectories (section 4.4).

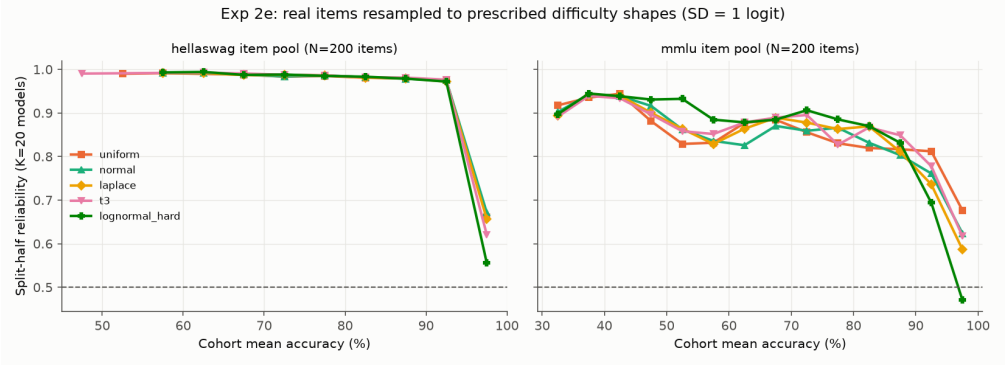


Figure 2: **Real items resampled into prescribed difficulty shapes** (200 items, SD 1 logit, $K = 20$ held-out real LLMs). *(Left)* HELLASWAG pool: all shapes are near-perfectly reliable up to 93% accuracy and drop together above 95%. *(Right)* MMLU pool: shapes differ by less than replicate noise below 90%; above 95%, uniform holds best (0.68) and the heavy hard tail is the only shape to fall below 0.5.

		Cohort mean accuracy			
Item pool	**Shape**	80–85%	85–90%	90–95%	95–100%
MMLU	uniform	0.82	0.82	**0.81**	**0.68**
	normal	0.83	0.80	0.76	0.62
	Laplace	**0.87**	0.80	0.74	0.59
	t_3 (heavy tails)	**0.87**	**0.85**	0.78	0.62
	lognormal-hard (heavy hard tail)	**0.87**	0.82	0.69	0.47
HELLASWAG	all five shapes	0.98	0.98	0.97	0.55–0.62

Table 2: **Direct test of H1 with real items.** Mean split-half reliability of 200-item benchmarks resampled from real MMLU or HELLASWAG items into prescribed difficulty shapes (SD 1 logit) and scored with real responses of $K = 20$ held-out models. Best per column in **bold**. Uniform does *not* saturate at 85%; it is the most robust shape above 90% accuracy on MMLU, and the heavy hard tail is the least robust.

4.1 H1: Difficulty shape barely moves the saturation point

In the controlled model, all shapes saturate together. Figure 1 shows reliability and Kendall τ as the cohort moves toward the ceiling. The seven curves are nearly indistinguishable until about 90% accuracy and cross $\rho = 0.5$ within a narrow window. Table 1 gives the saturation accuracy $A^\star$: in the base setting it ranges from 95.2% to 96.6%. Using Kendall $\tau < 0.5$ instead gives $A^\star = 95.7$–96.9% for every shape. The ordering is the opposite of H1. Uniform is among the latest to saturate (96.6%), and the shape with a long hard tail (lognormal-hard) is the *first* to saturate in every setting, by 1–4 points.

Item count and cohort spread move the saturation point by 15–20 points. The rows of table 1 vary one factor at a time. Cutting the benchmark to 100 items lowers $A^\star$ to 79–85%, and growing it to 2,000 items raises it to 98.5–99%. Halving the cohort spread to $\sigma_c = 0.15$ lowers $A^\star$ to 83–88%, and doubling it to 0.6 raises it to 98.5–99%. The 85%-vs.-95% contrast predicted by H1 therefore does appear, but along the axes of benchmark size and cohort tightness, not shape. Adding a 25% guessing floor changes $A^\star$ by at most 0.4 points. Only when the difficulty distribution is very wide (SD $= 3$) do shapes spread out more (92.4–96.7%), and again the hard tail is worst.

Shape does affect lifetime, but in the opposite direction. Measured as the length of the ability axis with $\rho \geq 0.5$, uniform has the longest lifetime among matched-variance shapes: 9.0 logits in the base setting vs. 8.2 for lognormal-hard, and 12.5 vs. 9.2 at SD $= 3$ (table 6 in the appendix).

Real items reproduce the null. Figure 2 and table 2 test H1 directly: we build 200-item benchmarks from real MMLU and HELLASWAG items with prescribed difficulty shapes and score them with real model responses. On MMLU, the uniform benchmark keeps $\rho = 0.81$ at 90–95% accuracy

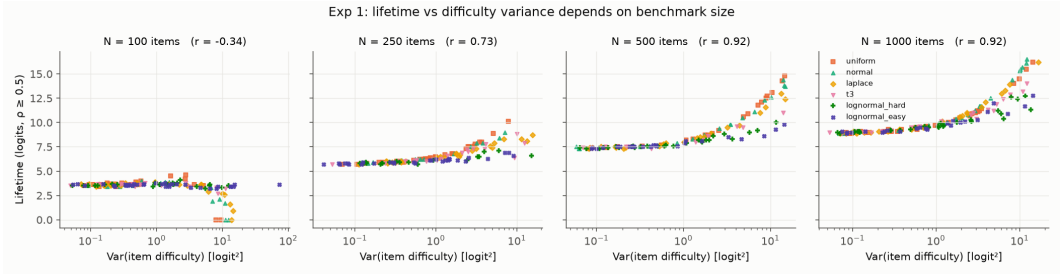


Figure 3: **Lifetime vs. difficulty variance in 600 random IRT benchmarks, split by size.** Lifetime is the length of the ability axis (logits) with $\rho \geq 0.5$. At $N \geq 500$, more variance means a longer lifetime ($r = 0.92$). At $N = 100$, the relation reverses ($r = -0.34$): spreading few items over a wide range leaves too few near any one ability level, and some wide 100-item benchmarks never reach $\rho \geq 0.5$. Pooled over sizes, $r = 0.20$.

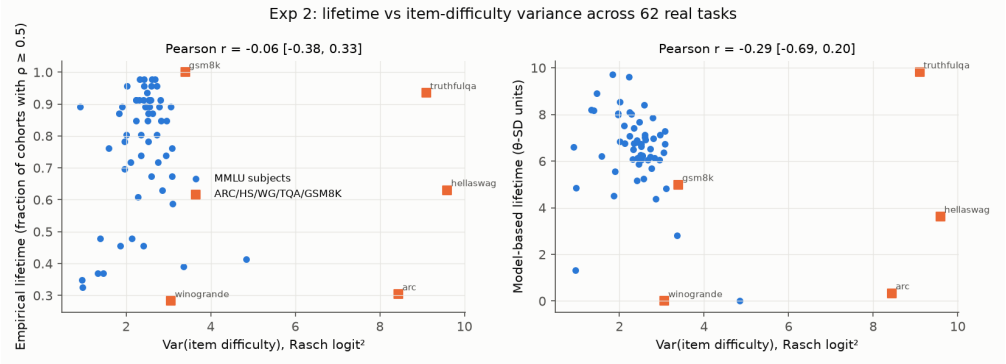


Figure 4: **Lifetime vs. item-difficulty variance across 62 real tasks.** *(Left)* Empirical lifetime (fraction of 20-model cohorts with split-half $\rho \geq 0.5$ on 90-item subsamples). *(Right)* Model-based lifetime from each task's fitted Rasch difficulties. Blue: MMLU subjects; orange: the five other benchmarks. Neither panel shows a positive relation. HELLASWAG has the largest $\mathrm{Var}(b)$ but a short lifetime; GSM8K has modest $\mathrm{Var}(b)$ but stays discriminative throughout.

and 0.68 above 95%; it is the most robust shape near the ceiling, not the least. The hard-tailed benchmark is the only one that drops below 0.5 (0.47 above 95%). On HELLASWAG, all five shapes stay at $\rho \geq 0.97$ up to 95% and fall together to 0.55–0.62 above it. Neither pool shows uniform benchmarks failing at 85%.

4.2 H3: Difficulty variance predicts lifetime only at fixed benchmark size

In the IRT model, the correlation depends on size. Figure 3 plots lifetime against $\mathrm{Var}(b)$ for 600 random benchmark configurations. Within a fixed size, the relation is strong for large benchmarks: $r = 0.92$ at both $N = 500$ and $N = 1000$, and the 90% difficulty range predicts lifetime even better ($r = 0.96$ at $N = 500$). It weakens to $r = 0.73$ at $N = 250$ and reverses to $r = -0.34$ at $N = 100$. Pooled across sizes, the correlation is only $r = 0.20$ (Spearman 0.27), because size differences dominate lifetime. H3 thus holds in theory, but only within a family of equal-size benchmarks with at least 500 items.

Across real tasks, the correlation vanishes. Table 3 and figure 4 report the same test on 62 real tasks. Empirical lifetime is uncorrelated with $\mathrm{Var}(b)$ ($r = -0.06$, 95% CI $[-0.38, 0.33]$), and the model-based lifetime is, if anything, negatively correlated ($r = -0.29$). Controlling for mean accuracy does not change this, and neither do alternative measures of spread ($|r| < 0.3$ for all). Restricting to the 57 MMLU subjects, which share one format, gives $r = 0.18$ ($p = 0.17$). Two tasks illustrate the gap between variance and lifetime: HELLASWAG has the largest difficulty variance yet loses discrimination for the top cohort, whereas GSM8K has modest variance yet stays discriminative for every cohort.

Lifetime measure	Pearson r [95% CI]	Spearman	Partial r | mean acc.
Empirical (fraction of cohorts with split-half $\rho \geq 0.5$)	-0.06 $[-0.38, 0.33]$	0.08	-0.14
Empirical (binomial ρ)	-0.10 $[-0.45, 0.30]$	-0.17	-0.08
Model-based (real difficulties, θ-SD units)	-0.29 $[-0.69, 0.20]$	-0.35	-0.25

Table 3: **Lifetime vs. item-difficulty variance** $\mathrm{Var}(b)$ **across 62 real tasks** (220 leaderboard models). Confidence intervals are from 2,000 bootstrap resamples. No measure comes close to the hypothesised $r > 0.8$; every interval includes zero. Other variance measures (SD, IQR, 90% range, $\mathrm{Var}(\mathrm{logit}\, p)$) all give $|r| < 0.3$.

			Seed share		Ranking stability (greedy)	
Benchmark	**Cohort**	**Mean acc.**	**greedy**	**sampled**	**Kendall τ**	**Pair flip prob.**
MMLU	all 15	60%	0.02	0.02	0.95	0.03
MMLU	top 6	69%	0.10	0.10	0.90	0.06
ARC-CHALLENGE	all 15	78%	0.01	0.01	0.95	0.04
ARC-CHALLENGE	top 6	88%	0.10	0.09	0.89	0.09
ARC-EASY	all 15	90%	0.01	0.01	0.91	0.06
ARC-EASY	top 6	96.0%	**0.18**	**0.15**	0.81	0.14

Table 4: **Seed noise on natural 500-item benchmarks (15 open LLMs, 10 option-order seeds).** Seed share is the fraction of score variance due to evaluation seeds; Kendall τ compares a single-seed ranking with the seed-averaged ranking; pair flip probability is the chance that two single-seed runs order a model pair differently. Even for the near-ceiling ARC-EASY top-6 cohort (96% accuracy), seeds explain only 18% of score variance (**bold**: the largest share). Item-bootstrap upper bounds of the seed share stay below 0.5 for every row (largest: 0.46, ARC-EASY top 6).

Real cohorts saturate on the most saturated tasks. On 90-item subsamples, real 20-model cohorts rarely exceed a split-half reliability of 0.7 (figure 8 in the appendix). On the two most saturated tasks the top cohort loses discrimination: HELLASWAG drops from a peak of 0.96 for lower cohorts to 0.10 for the top cohort (91% accuracy), and WINOGRANDE from 0.77 to 0.23 (86% accuracy). In contrast, GSM8K (generative) and TRUTHFULQA keep high reliability for the top cohort (0.72 and 0.91).

4.3 H2: Seed noise dominates only for near-tied models on small, near-ceiling test sets

Seed noise is a fixed fraction of item-sampling noise. Seed-averaged greedy accuracy on ARC-EASY ranges from 59.5% (Qwen2.5-0.5B) to 97.3% (Qwen2.5-14B). Across models, the median seed SD of accuracy on 500 items is 0.75 points on ARC-EASY (range 0.43–1.83), 0.95 on ARC-CHALLENGE (0.43–2.22), and 1.30 on MMLU (0.68–2.54). Seed variance scales like binomial variance: $N \cdot \mathrm{Var}_{\text{seed}}(\text{acc}) \approx 0.40 \times \text{acc}(1 - \text{acc})$ (median ratio; 0.40 on ARC-CHALLENGE, 0.45 on ARC-EASY, 0.37 on MMLU). Option-order seeds therefore add noise equal to roughly 40% of the noise from sampling the items themselves.

On full 500-item benchmarks, seeds explain at most 18% of variance. Table 4 shows that seed share grows as the cohort narrows and accuracy rises, from 1–2% for the full 15-model cohorts to 10% for the top 6 on MMLU and ARC-CHALLENGE, and 18% for the ARC-EASY top 6 at 96% accuracy. Rankings are correspondingly stable (single-seed Kendall $\tau \geq 0.81$). The top-1 model is the least stable quantity: across settings, a single seed picks the seed-averaged winner in 40–100% of cases, and on ARC-EASY a single greedy seed finds it only 40% of the time. Sampling at temperature 1 adds little beyond option order (seed share 0.15 vs. 0.18).

Seed share rises with accuracy but crosses 0.5 only at the very top. Figure 5 uses difficulty-stratified 150-item subsets. Seed share rises monotonically with cohort accuracy (pooled Spearman $\rho = 0.96$, $p < 10^{-8}$ greedy; 0.97, $p < 10^{-9}$ sampled). The mean share is 0.46 at accuracy $\geq 90\%$ vs. 0.15 below 80% (greedy). On MMLU it peaks at 0.42 (83% accuracy). On ARC-CHALLENGE and ARC-EASY it exceeds 0.5 only at $\geq 99.2\%$ accuracy. H2 thus has the right direction but requires a much stronger form of saturation than "mean accuracy above 90%".

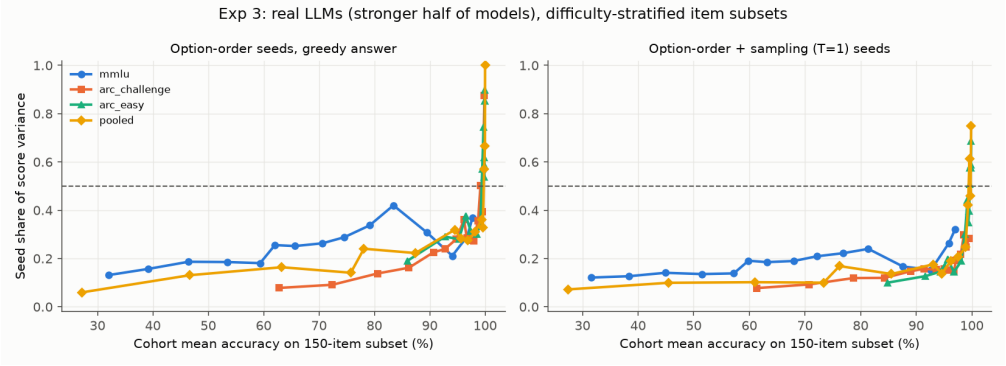


Figure 5: **Seed share of score variance vs. cohort accuracy on difficulty-stratified 150-item subsets** (stronger 8 of 15 LLMs; difficulty estimated on the weaker 7). *(Left)* Option-order seeds, greedy answers. *(Right)* Option order plus sampling at $T = 1$. Seed share rises steadily with accuracy (Spearman 0.96 greedy, 0.97 sampled) but crosses 0.5 (dashed) only at $\geq 99\%$ accuracy on ARC-CHALLENGE and ARC-EASY, where nearly every model solves nearly every item.

Benchmark	Top-20 mean	Top-20 SD	Pred. seed SD	Pred. seed share
ARC	76.6%	2.6 pts	0.78 pts	0.09
HELLASWAG	89.4%	1.7 pts	0.19 pts	0.01
MMLU	78.1%	1.1 pts	0.22 pts	0.04
TRUTHFULQA	73.6%	5.0 pts	0.97 pts	0.04
WINOGRANDE	85.5%	1.6 pts	0.62 pts	**0.14**
GSM8K	79.7%	3.8 pts	0.70 pts	0.03

Table 5: **Predicted seed share for the real Open LLM Leaderboard v1 top-20 (June 2024).** We apply the seed-noise rate fitted in Exp 3 ($N \cdot \mathrm{Var}_{\mathrm{seed}} \approx 0.40\,\mathrm{acc}(1-\mathrm{acc})$) to each benchmark's size and top-20 score spread. The largest predicted share (**bold**) is 14%, for the smallest benchmark (WINOGRANDE, 1,267 items).

Item count again decides. For the ARC-EASY top-6 cohort, the seed share falls from 0.51 at 50 items to 0.45 at 100, 0.32 at 200, and 0.18 at 500 items (table 7 in the appendix). Seed noise shrinks as $1/N$ while the capability gap does not, so large benchmarks rarely let seeds dominate.

Real leaderboards are far from the seed-dominated regime. Table 5 applies the fitted seed-noise rate to the real Open LLM Leaderboard v1 top-20 as of June 2024. Predicted seed shares range from 1% (HELLASWAG) to 14% (WINOGRANDE). With 800–14,000 items per benchmark, seed noise is small compared with the spread of the frontier, even on benchmarks above 85% mean accuracy.

4.4 Real leaderboard trajectories

Figure 6 shows how the real frontier evolved. Frontier means rose on every benchmark, fastest on GSM8K (31% to 80%). Reliability against item-sampling noise stayed at or above 0.9 for HELLASWAG, MMLU, GSM8K and TRUTHFULQA in most months after January 2024; HELLASWAG and MMLU have 10k–14k items. The smaller benchmarks were less reliable even at lower accuracy: WINOGRANDE (1,267 items) at about 0.64 and ARC at about 0.4–0.8. Most strikingly, in some months (e.g., December 2023) $\rho \approx 0$ on *every* benchmark at once. In these months the top 20 was dominated by near-duplicate merged models whose scores differed by less than item-sampling noise. This is saturation of the *cohort*, not of any benchmark, and it confirms that cohort spread is as important as benchmark design in deciding whether ranks are meaningful.

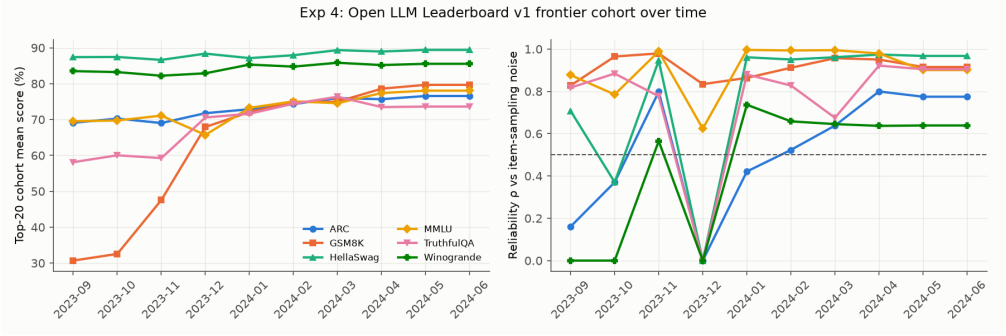


Figure 6: **The Open LLM Leaderboard v1 frontier cohort (top 20 unique models) by month.** *(Left)* Cohort mean score. GSM8K rose fastest (31% to 80%), MMLU rose from 70% to 78%, and HELLASWAG from 87% to 89%. *(Right)* Reliability ρ against item-sampling noise. Large benchmarks stay above 0.9 in most months after January 2024; smaller ones (WINOGRANDE, ARC) are less reliable. In December 2023, $\rho \approx 0$ for every benchmark at once because near-duplicate model merges dominated the top 20.

5 Discussion

5.1 Why shape matters so little: a back-of-the-envelope account

A short calculation explains the three main findings. Consider the Rasch case of equation 2 ($a_i = 1$, $c = 0$) and a tight cohort, so we can linearise scores around the cohort centre μ. Write $v = \frac{1}{N}\sum_i P_i(1 - P_i)$ for the mean item variance at μ. The slope of expected accuracy in ability is $\partial\bar{s}/\partial\theta = v$, so the signal is $\mathrm{Var}_k(\bar{s}_k) \approx v^2\sigma_c^2$, while binomial noise is about v/N. Substituting into equation 1 gives

$$\rho \approx \frac{Nv\sigma_c^2}{Nv\sigma_c^2 + 1}, \qquad \text{so} \qquad \rho < 0.5 \iff v < \frac{1}{N\sigma_c^2}. \tag{3}$$

This approximation is ours; we use it only to interpret the exact computations above.

Why does shape matter little? Because $x(1 - x)$ is concave and $x(1 - x) \leq 1 - x$, the mean item variance is capped by the error mass: $v \leq A(1 - A) \leq 1 - A$, with equality on the left only when all items are equally hard. At a fixed accuracy, no difficulty shape can create more informative items than the remaining errors allow. Shape only decides how much of this capped budget is used. Shapes without a long hard tail keep the errors on items that cohort models solve part of the time, so v stays close to its cap. A heavy hard tail concentrates errors on items that no model in the cohort solves ($P_i \approx 0$), which contribute almost nothing to v. This is exactly the ordering we observe: the hard-tailed shape saturates first (table 1, table 2).

Why do size and cohort spread dominate? The bound $v \leq 1 - A$ and equation 3 imply that discrimination is lost no later than $A^\star \leq 1 - 1/(N\sigma_c^2)$. This gives 97.8% for the base setting, 88.9% at $N = 100$, 99.4% at $N = 2000$, 91.1% at $\sigma_c = 0.15$, and 99.4% at $\sigma_c = 0.6$. The exact $A^\star$ values in table 1 lie below this bound in every case and follow its ordering. Both N and σ_c^2 enter as a product, which is why a tight cohort and a small benchmark have the same effect.

Why does variance help only large benchmarks? Wider difficulties extend the range of abilities for which some items sit near the cohort, which lengthens lifetime when N is large enough that the local density stays above $1/(N\sigma_c^2)$. With few items, spreading them thins the density everywhere, so v never gets large enough, and lifetime shrinks (figure 3, $N = 100$).

When does seed noise dominate? Adding seed noise of $r\,v/N$ with $r \approx 0.40$ (Exp 3), the seed share becomes $r/(r + Nv\sigma_c^2)$, which exceeds 0.5 only when $Nv\sigma_c^2 < r$, a stricter condition than the $\rho < 0.5$ condition in equation 3. This matches our measurements qualitatively: seeds dominate only for tight cohorts on small subsets at $\geq 99\%$ accuracy, and the seed share falls as N grows (0.51 at 50 items to 0.18 at 500 for the ARC-EASY top 6), though more slowly than the $1/N$ scaling of this approximation would suggest.

5.2 Implications

For benchmark builders. The most useful lever is *item informativeness near the frontier*: the number of items, weighted by their discrimination, whose difficulty lies close to the ability of the models that will be compared. A wide or heavy-tailed histogram does not provide this automatically; a hard tail spends items on questions no current model can answer. Builders should keep N large, budget items near the expected frontier (e.g., with an adaptive or IRT-targeted item bank [Zhuang et al., 2023, Maia Polo et al., 2024]), and refresh the bank as the frontier moves.

For leaderboard readers. On today's large benchmarks, rank differences are rarely seed artefacts: predicted seed shares for the Open LLM Leaderboard v1 top-20 are at most 14%. The exception is near-tied frontier models on small test sets. When a test set has ≤ 200 items, accuracy is above 95%, and the top models are within 1–2 points, rank differences should be reported with seed-level and item-level error bars [Miller, 2024]. Even then, the top-1 position is fragile: a single seed recovered the seed-averaged winner only 40% of the time on ARC-EASY.

For leaderboard maintainers. Several months of the Open LLM Leaderboard v1 had $\rho \approx 0$ on every benchmark because near-duplicate merges filled the top 20. Deduplicating cohorts by base model would remove this form of saturation, which no benchmark design can fix.

5.3 Surprises

Three findings were unexpected. First, 26% of ARC-CHALLENGE items are answered below chance by the model population, presumably because of adversarial distractors; this breaks standard guessing-floor IRT models, and we used a Rasch model instead. Second, as noted above, leaderboard top-20s often saturate because of merges rather than the benchmark itself. Third, Qwen3-8B, Phi-3.5 and SmolLM2 put only 12–33% of next-token probability on the bare answer letters. Renormalised accuracies are still sensible (e.g., Qwen3-8B reaches 91% on ARC-CHALLENGE), but this adds format-sensitivity noise on top of option order.

5.4 Limitations

Seed type. Our seeds are *evaluation* seeds (option order and sampling). Training-seed variance, as measured by Madaan et al. [2024], could not be measured here and may be larger, which would lower the accuracy at which seeds dominate.

Model cohort. Exp 3 uses 15 open models of 0.5–14B parameters rather than frontier API models; no API keys were available, and two gated model families (Llama-3.1-8B, Gemma-2-9B) were not accessible. ARC and MMLU are likely in these models' training data, which affects absolute accuracy but not the variance decomposition.

Leaderboard data. Exp 2 and 4 use Open LLM Leaderboard v1 data (open models from 2023–24, log-probability scoring), which saturated relatively early; per-item v2 data is gated. The ability axis in Exp 2 is an external composite rather than calendar time. Calendar-time lifetimes were measured only for six benchmarks (Exp 4) and are confounded by cohort composition.

Model fit and resampling. The Rasch model is unidimensional; pooling MMLU subjects and per-task difficulty variances carries multidimensionality and prior-shrinkage effects, and extreme items are bounded by the $\mathrm{Normal}(0, 3^2)$ prior. Importance resampling from finite pools only approximates target shapes (e.g., the achieved kurtosis was 4–5 for t_3 and 10–16 for lognormal-hard). Item-bootstrap intervals for the seed share are biased upward, because duplicated items inflate seed variance; this makes our "below 0.5" conclusions conservative.

Broader impact. Our results are meant to make benchmark rankings more trustworthy. A possible misuse is to dismiss real capability gaps as "noise"; our measurements show that on large benchmarks this is rarely justified.

6 Conclusion

We asked whether the shape of a benchmark's item-difficulty distribution decides when it stops ranking models, and whether evaluation-seed noise dominates rankings at that point. Across a matched-variance IRT model, 220 leaderboard models on 62 real tasks, real items resampled into prescribed

shapes, and 15 open LLMs evaluated under 10 seeds, the answer is mostly no. All difficulty shapes lose discrimination within about 1.5 points of each other, and a heavy hard tail saturates first, not last. The saturation point is set mainly by the number of items (80–84% accuracy at 100 items, 96% at 500, 99% at 2000) and by how close the competing models are. Difficulty variance predicts lifetime only among equal-size benchmarks ($r = 0.92$ in simulation) and not across real tasks ($r = -0.06$). Seed noise is about 40% of item-sampling noise and dominates only for near-tied models on small test sets at $\geq 99\%$ accuracy.

The key takeaway is simple: **to keep a benchmark useful, add informative items near the frontier and keep it large; the shape of its difficulty histogram is a second-order concern.**

Future work should repeat the seed analysis with frontier API models and training-seed replicates, measure lifetimes in calendar time using per-item data with release dates (e.g., leaderboard v2 or HELM [Liang et al., 2023]), test the summed Fisher information at the cohort's ability as a direct lifetime predictor, and study how deduplicating leaderboard cohorts by base model changes apparent saturation.

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A Additional Results

Setting	point	uniform	normal	Laplace	t_3	lognormal-hard
Base (SD $= 1.5$, $N = 500$)	7.3	**9.0**	**9.0**	8.9	8.5	8.2
SD $= 3.0$	7.3	**12.5**	12.3	11.3	10.4	9.2
$N = 100$	3.5	**4.3**	4.0	3.8	3.8	3.6

Table 6: **Effective lifetime in the IRT model** (logits of frontier ability with $\rho \geq 0.5$). Longest lifetime per row in **bold**. Uniform has the longest lifetime among matched-variance shapes; the heavy hard tail is shorter. The point mass does not depend on SD.

Items	50	100	200	500
Seed share (greedy)	0.51	0.45	0.32	0.18

Table 7: **Seed share vs. benchmark size** for the ARC-EASY top-6 cohort (96% accuracy). Seed noise dominates only at 50 items.

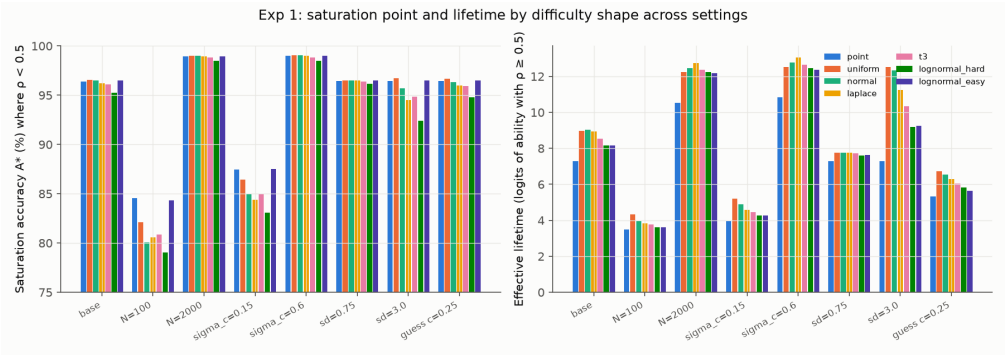


Figure 7: **Saturation accuracy $A^\star$ *(Left)* and effective lifetime *(Right)* for each difficulty shape across sensitivity settings** of the IRT model. Within each setting, bars differ little in $A^\star$; across settings, item count (N) and cohort spread (σ_c) produce the large changes.

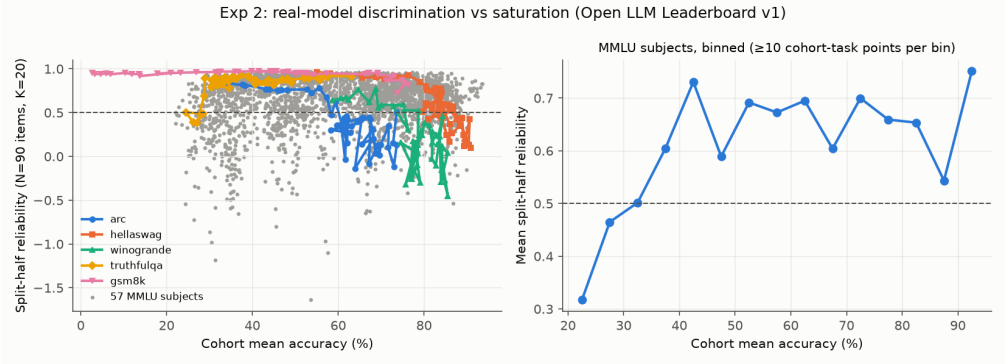


Figure 8: **Real-model discrimination vs. cohort accuracy** (Open LLM Leaderboard v1, 20-model cohorts, 90-item subsamples). *(Left)* Split-half reliability per task; grey points are the 57 MMLU subjects. HELLASWAG and WINOGRANDE lose discrimination for the top cohorts, while GSM8K and TRUTHFULQA stay reliable. *(Right)* Mean reliability of MMLU subjects binned by accuracy; it does not decline toward high accuracy over the observed range.

B Experimental Details

IRT model (Exp 1). Discriminations $a_i \sim \mathrm{LogNormal}(0, 0.3^2)$. Difficulty shapes are rescaled to the target SD and centred. The cohort centre is swept over 221 ability levels; accuracy mean and variance are computed exactly with the Poisson-binomial distribution. Kendall τ uses 200 Monte Carlo replicates per level. Each shape is replicated over 20 independent item draws.

Leaderboard data (Exp 2, Exp 4). Data come from the public `open-llm-leaderboard-old` details and contents datasets. Of 245 sampled models, 227 had at least 60 of 62 tasks and 220 had all 62. Metrics: ARC-CHALLENGE acc_norm (25-shot), HELLASWAG acc_norm (10-shot), TRUTHFULQA mc1 (0-shot), WINOGRANDE acc (5-shot), GSM8K exact match (5-shot), MMLU acc (5-shot). The Rasch fit uses joint MAP estimation with L-BFGS in PyTorch.

Seed evaluation (Exp 3). Items: 500 test items each from ARC-EASY, ARC-CHALLENGE [Clark et al., 2018] and MMLU [Hendrycks et al., 2021], sampled with seed 42. Models were run in bf16 with Hugging Face `transformers`. Seed 0 uses the original option order; seeds 1–9 apply random permutations. The answer distribution is the next-token probability over letters A–D after the prefix "The answer is", renormalised. Total: 15 models $\times$ 1,500 items $\times$ 10 seeds $=$ 225,000 forward passes.